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Multi-echelon and multi-period supply chain management network design considering different importance for customers management using a novel meta-heuristic algorithm

Esra Sipahi Döngül[a,*], Erkin Artantaş[b], Mutlu Başaran Öztürk[c]

[a] Faculty of Health Science, Department of Social Work, Aksaray, Turkey
[b] Osmaniye Korkut Ata University, Turkey
[c] Nigde Ömer Halisdemir University, F.E.A.S, Turkey

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ABSTRACT

Supply chain management is an integrated approach to planning and controlling materials and information from suppliers and flows to customers. Given the strategic planning of distribution networks, decisions about it must be optimized. Among the needs for optimization in the distribution network are facility location decisions and inventory control decisions. In this paper, an integrated location-allocation model with inventory control decisions is presented by considering the importance of customers. The proposed model is proposed for a multi-echelon and multi- period supply chain, in which the demand is considered as an uncertain parameter. The main innovation of this research is to consider the distinction between customers because customers with different purchasing volumes in conditions of demand uncertainty have a different level of importance for the supply chain. Moreover, a novel meta-heuristic algorithm as Seeker Evolutionary Algorithm (SEA) is applied to solve the proposed model. The results of comparing SEA with genetic algorithm and GAMS software show that both in small and large dimensions, SEA provides high-quality solutions, and its speed of operation is at an acceptable level.

1. Introduction

In today's competitive market, with an uncertain and risky operating environment, designing a supply chain network in a cost-effective and responsive manner is crucial (Biuki et al., 2020). Understanding how the various criteria for evaluating suppliers play a role is crucial for a better understanding of the dynamics of the supply chain Pani & Kar (2011). In the 1960s and 1970s, organizations sought to increase their competitiveness by standardizing and improving their internal processes to produce high quality and low-cost products. The fundamental thinking at the time was that solid engineering and design, as well as coherent production operations, were prerequisites for achieving market demands and thus gaining more market share. Accordingly, supply chains focused their efforts on increasing efficiency (Sebatjane & Adetunji, 2021).

In the 1990s, along with improvements in production processes and the application of reengineering models, managers in many industries realized that improved internal processes and flexibility in the company's capabilities were sufficient to maintain market presence and parts and materials suppliers. They must produce materials with the best quality and the lowest cost. Product distributors should also be closely related to producer market development policies (Goli & Malmir, 2020). Accordingly, the production industries attended to prepare different supply chains. The supply chain is a network of facilities (e.g. manufacturing plants, distribution centers, warehouses, etc.) that operate from the supply of raw materials to their conversion into intermediate or final products and finally the distribution of these products to the customers (Vicil & Jackson, 2006). In other words, the supply chain is created from all parties involved - directly or indirectly in meeting customer demand.

The supply chain includes manufacturing and supply activities and the activities of transporting goods, warehousing, retailers, and even customers. Supply chain management decisions are classified into three levels: strategic, tactical and operational (Sangaiah et al., 2020; Vicil & Jackson, 2006). A study by Kar & Pani (2014) aimed to identify critical supplier selection criteria that are important across industries and purchasing contexts, and then estimate the importance of these criteria for procurement practitioners. Kar & Pani (2014) analyzed data from 188 companies in 12 sectors with a murky analytical hierarchy process to determine critical supplier selection criteria in Indian manufacturing industries. As a result of the research, they found that seven criteria are critical for Indian manufacturing industries. In addition, while evaluation criteria such as product quality, delivery compliance and price have maximum criticality, they have found that criteria such as

* Corresponding author.
E-mail address: esrasipahidongul@aksaray.edu.tr (E.S. Döngül).

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e-processing capability have gained importance with the spread of e-tender platforms.

Facility Location Problems is one of the most strategic decisions in the supply chain in such a way that the lowest cost to meet customer demand depends on it (Goli et al., 2020). On the other hand, different customers have different values for the supply chain and based on it, they need different levels of service depending on their level of importance. In this regards, (Melo et al., 2009) introduced the following scenarios:

1 Serve all demand with a common safety stock. This option provides over-service to all customers with lower than maximum performance, which indicates a waste of resources.
2 Use a special confidence reserve for each class of customers. This option does not have the ability to accumulate maximum inventory and tends to maintain maximum reliability.
3 Provide demand from joint accumulation but using a quota policy. In this scenario, as soon as inventory falls below a certain critical level, the service to low-demand requests is stopped and the delivery of these requests is delayed until the next order arrives. Here inventory on hand and deferred orders for low priority customers at the same time may be positive. Compared to the several research items done in the field of inventory optimization, few ones have considered inventory with different customer classes (Melo et al., 2009). Perhaps the first published study to consider different customer classifications was presented by Tempelmeier (2006), who introduced the critical level concept with a dynamic system by reviewing cycles. This approach has been proven as an optimal policy by Veinott (1965) for the issue of missed sales such as deferred orders. This is also mentioned in Topkis (1968). Wang et al. (2013) considered two classes of customer as high-priority L-class customers and low-priority B-class customers due to the lack of order, delayed and lost sales at the same time. Zhou & Zhao (2010) proposed a two-tier inventory system in the supply chain in which the mathematical modeling can determine the optimal level of inventory in the entire supply chain network Abdolazimi et al. (2021) studied the determination of the optimal inventory number of each product group in a supply chain. ABC analysis was used in this study. In this regard, a multi-objective model was proposed. The first objective was to maximize supply chain profits and the second was to increase the service level. In order to optimize this model, NSGA-II and MOPSO algorithms was used Kar (2014) proposes an approach for group decision support for the supplier selection problem by integrating fuzzy Analytic Hierarchy Process (AHP) for group decision making and fuzzy goal programming for discriminant analysis. Young & Steele (2022) says it's hitting ampric analysis. They measured the performance of predictive models that predicted eye condition and emergency patient mortality from a broad dataset of more than 1.83 million patients discharged. Both two and a half years of relatively slow performance application and performance-applied speed between machine training model types and the training set's time period achieve significant empirical results up to the end-of-scope points.

Recently, Omar et al. (2020) studied Vendor Managed Inventory (VMI) in the supply chain and examined the application of Blockchain technology. For this purpose, a public framework was introduced, and artificial intelligence algorithms were used to evaluate it. Artificial intelligence's ability to enhance the applicant and employee involvement by automating routine, low-value responsibilities, and freeing up time to concentrate on the more planned, innovative work that teams need and want to do has been a burning topic in the research world for years (Sipahi & Artantas, 2022). Deepu & Ravi (2021) research has introduced a new model for choosing the best IOIS alternative, taking into account the content, scope, and critical decision-making factors that affect supply chain integration. They identified twelve decision-making factors that influenced IOIS selection and categorized them under four technologically important dimensions: implementation and innovation for operational, effective decision-making. They concluded that project completion time is the most relevant criterion and follows the financial resources required to select digital technology providers and IOIS alternatives.

Konstantakopoulos et al. (2021) identified a situation in which there was a fleet and resources workflow in logistics route and scheduling operations in a study. By calculating the state status, implied state, and the data daily distribution statuses are calculated independently of the rows, which predict indirect and cost reductions. This daily case, calculated by "Third Party Logistics" in Greece, is based on what work today is done separately or in some way in a heuristic way within the values of the advantages of the common case.

Another study by Sarkar & Shankar (2021) worked and analyzed barriers to port logistics during industry 4.0 for emerging economies. A total of eight key barriers were selected from the literature. Barriers include a TISM for a modular model that reveals between connections to management. Apply MICMAC analysis to categories in clusters, depending on the strength and propulsion. Author said unsymed politics, weak infrastructure and R&D barriers were barriers.

Yoo & Cheong (2021) examined the stability of the closed-loop supply chain based on inventory behavior. For this purpose, a mathematical model is presented that tries to maximize the whole chain's profit and determine the optimal amount of inventory at different levels. There is also a network interface to try to increase the stability level of supply chain. By reviewing various research items, the innovation of this research can be summarized listed as follows:

- Incorporating the location and inventory optimization in the supply chain due to demand fluctuations.
- Investigating of the importance of customers and their grouping in inventory plan in the supply chain.
- Considering the critical levels for supply chain inventory management and product allocation for customers.
- Implementation of Seeker Evolutionary Algorithm as a new and efficient meta-heuristic algorithm in the field of supply chain network design.

Accordingly, in this research, location-inventory is focused on the supply chain network design problem. The studded supply chain includes Production systems, ware-houses and consumers. The purpose of this study is to determine the amount of product flow in the supply network, locating warehouses among potential locations, and meet consumer needs in a three-echelon supply chain. Moreover, determining the level of inventories by considering the differentiation of customers along the planning horizon in the supply chain is one of key issues in modeling and optimizing the supply chain network design.

The rest of the paper is arranged as follows. In Section 2, the problem statement and mathematical modeling are presented. In Section 3, the solution method of the mathematical model is specified and in Section 4, the results of optimizing the mathematical model are presented. Finally, Section 5 summarizes the research and provides the future outlooks.

2. Problem statement and mathematical model

Inspired by Abdolazimi et al. (2021), Kaplan (1969), Yoo & Cheong (2021), Zhou & Zhao (2010), the studded supply chain is multi-echelon and responsible for meeting customers' demand. The produced stocks are not directly available to the customer and after being stored in warehouses, they are sent according to the customers' demand. In addition to determining inventory levels, the model determines the amount of confidence reserves for each customer category. Accordingly, an integrated supply chain network design is presented, in which inventory and location decisions are optimized simultaneously.

Kar & Pani (2014) applied group decision support theories for supplier selection and were used to prioritize group preferences under consensus in the context of the line geometric mean method (RGMM) of

the analytical hierarchy process (AHP). The agreement-based group conducted a case study to test decision-making theories and compare them with other AHP-based approaches. Findings further imply that RGMM is more effective than eigen value method, for group decision making under consensus.

In this study, the capacity limit for warehouses is different, and distribution centers' capacity limits the allocation of customers to distribution centers. In the term of customer grouping, for each series of customers, when the inventory level is less than the critical level, the customer demand will be delayed and penalties will be considered for it in the objective function. In this model, each customer has an indefinite demand. The objective function of this model is to select a set of warehouses (distribution centers) to serve customers and allocate customers to opened distribution centers so that the total inventory, location and shortage for each series of customers is minimized.

The critical level for the total customer is the level of inventory that if the total inventory level falls below this value, the total customer will no longer be answered and until refilling, the relevant customer order will be considered as a delayed order. For example, if the critical level of second-class customers per product is 50 ($C2 = 50$), then as long as the stock is less than 50 units, customers who go to the warehouse will face with shortage in satisfying their demand. Therefore, the customers of this supply chain are classified so that important customers can be prioritized. These classes include Class B and Class L. Class B customers will be fined *Pb* if they do not respond to the lack of inventory. Moreover, a sharing policy (C,SS) with several critical levels and considering the shortfall as a deferred order is used. *Cj* is the critical level of customer class j for the product and SS is the base inventory level or the same as the confidence balance. There is also a relationship between critical levels and confidence reserves as follows (Yoo & Cheong, 2021).

$$C_1 \le C_2 \le \ldots \le Ck \le SS$$

Yoo & Cheong (2021) proved that the value of the variable C_1 at the optimal point is zero, which is obvious, because if we have inventory, we respond to customer demand with the highest priority. Fig. (1) shows the inventory manners in the proposed supply chain.

In the following of the paper, all items of the mathematical model are presented. After defining the indices, parameters and variables, the inventory and the shortage amount for each customer in each period is provided, and then the final model is obtained.

Indices

I: Index of customer

J: Index of potential warehouse locations

K: category of customers

Parameters:

TC_{ij}: The cost of sending each unit of product to customer *i* from the warehouse j

TC_j: The cost of sending each unit of product to warehouse *j* from the factory

F_i: Fixed cost to open a warehouse at location *j*

d_i: Average customer *i* demand for the product

v_i: Variance of customer *i* demand for the product

HC_j: Product storage cost in warehouse *j*

OC_j: Warehouse *j* order cost from the factory

Q_j: Order quantity of the *j*th product c_j. The price of *j* th product

B_j. Maximum backorder level of the *j*th product

cap_j: Potential storage capacity at location *j*

LT_j: The expected time to send the order from the factory to warehouse *j*

S: Product space required in each warehouse

PH: Planning horizon

$C1_j$ Critical level for customers with priority 1 in warehouse *j*

$C2_j$: Critical level for customers with priority 2 in warehouse *j*

Ck_j: critical level for customers with priority *k* in warehouse *j*

P^k: penalty for shortage of customers of degree *k*

TIC: total inventory costs

THC: total holding cost

TSC: total shortage costs

TSC^k: Cost of shortage of customers of degree *k*

TOC: Total order cost

TMC: Total purchase cost

TSCk: cost of shortage of customers of degree *k*

Variables:

Bj: The amount of product shortage in warehouse *j*

Xj: Binary variable and equal to 1 if the warehouse is established in location *j*

Yij: Binary variable and equal to 1 if warehouse *j* covers customer *i* demand

Dj: Average demand for product allocated to warehouse *j*

Vj: Variance of product demand allocated to warehouse *j*

Ckj: Critical levels of customer k in warehouse *j*

According to Fig. 1, Eqs. (1)–(7) are established to demonstrate the relation between different inventory variables.

$$T = \frac{Qj}{Dj} \tag{1}$$

$$t_{1=}t_{B=}\frac{Bj}{Dj} \tag{2}$$

$$t_3 = \frac{C3j}{Dj} \tag{3}$$

$$t_k = \frac{C4j}{Dj} \tag{4}$$

$$t_m = \frac{Qj - Ckj}{Dj} \tag{5}$$

$$t_2 = \frac{C2j}{Dj} \tag{6}$$

$$t_4 = \frac{C4j}{Dj} \tag{7}$$

where, *tm* is a period that supply chain responds to all customers and has no short-age, *t2* is when the inventory level is less than the critical level of customers in class 2, so during this time, customers are in shortage while the whole system has not been deficient and is also when the entire inventory system is deficient. Accordingly, Different costs of the supply chain can be calculated based on Eqs. (8)–(16)

$$\text{THC} = \frac{(Qj - Bj).(Tn)}{2}.\ HCj = \frac{(Qj - Bj).(Qj - Bj)}{2.Dj}.\ HCj \tag{8}$$

$$\text{Shortage of customer's in grade 1} = \frac{tB.Bj}{2} = \frac{B.B.J}{2Dj} \tag{9}$$

Shortage of customer's in grade 2

$$= \frac{(t2 + tB).\ (Bj + C2j)}{2} \tag{10}$$

Shortage of customer's in grade 3

$$= \frac{(t3 + tB).\ (Bj + C3j)}{2} \tag{11}$$

$$\frac{(Bj + C3j).(Bj + C3j)}{2Dj}$$

Shortage of customer's in grade k

$$= \frac{(Bj + Ckj).(Bj + Ckj)}{2Dj} \tag{12}$$

$$TSC = \left(\frac{B_j^2}{2.D_j}.P^1\right) + \left(\frac{(B_j + C_{2j}).(B_j + C_{2j})}{2.D_j}.P^2\right) + \left(\frac{(B_j + C_{kj}).(B_j + C_{kj})}{2.D_j}.P^k\right) \tag{13}$$

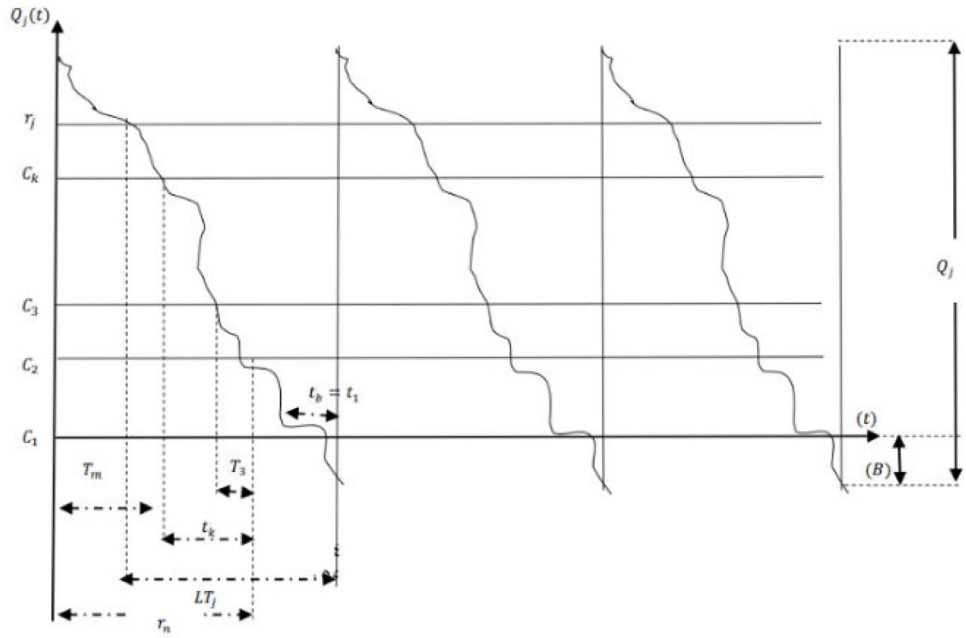

Fig. 1. The studded inventory style in the supply chain.

$$= \frac{\sum_{k=1}^{K}\left[(C_{kj}+B_j).(C_{kj}+B_j)\right].P^k}{2.D_j}$$

$$TOC = OC_j \tag{14}$$

$$\text{TIC} = N.(\text{Costofoneperiod}) \tag{15}$$

$$TIC = C_j \cdot D_j + \left(\frac{D_j}{Q_j} \cdot OC_j\right) + \left(\frac{(Q_j-B_j)\cdot(Q_j-B_j)}{2Q_j} \cdot HC_j\right) + \frac{\sum_{k=1}^{K}(C_{kj}+B_j)\cdot(C_{kj}+B_j)\cdot P^k}{2Q_j} \tag{16}$$

Now, to obtain Q_j^* and B_j^*, Eqs. (17), (18) are applied. The value of Q is given by Eq. (19) and the value of B is given by Eq. (20).

$$\frac{\partial TIC(Q_j, B_j)}{\partial(Q_j)} = 0 \tag{17}$$

$$\frac{\partial TIC(Q_j, B_j)}{\partial(B_j)} = 0 \tag{18}$$

$$Q^*_{j=2\sqrt{\frac{2D_j.\ OC_j+\ HC_j.\ B_j^2+\sum_{k=1}^{K}(C_{kj}+B_j).(C_{kj}+B_j).P^k}{HC_j}}} \tag{19}$$

$$B_j^* = \frac{Q_j^*.\ HC_j - \sum_{k=1}^{K} C_{kj}.P^k}{\left(\sum_{k=1}^{K} P^k\right) + HC_j} \tag{20}$$

Meanwhile, total cost is calculated according to the point of reordering and safety storage as Eq. (21).

$$TOC_{j=HC_j.Q_j/2+HC_j\ .Z_{1-a.}}\ \sqrt{LT_j} \cdot \sqrt{V_j} \tag{21}$$

Using the Qj * 209 value, TOC is updated based on Eq. (22).

$$TOC_j = HC_{j.}\left(2\sqrt{\frac{2Dj.OC_j + \sum_{k=1}^{K} + HC_{j.}B_j^2\left[(C_{kj}+B_j)\cdot(C_{kj}+B_j)\cdot p^k\right]}{HC_j}}\ /\ 2\right. + HC_{j.}Z_{1-a.}\sqrt{LT_j} \cdot \sqrt{V_j} \tag{22}$$

Therefore, the ultimate objective function of the problem, which is the total cost of warehouse establishment, holding, ordering, and shortage, will be in the form of Eq. (23), and the constraints of the mathematical model include Eqs. (24) to (32).

$$\min \sum_{j=1}^{J} Fj.Xj + \sum_{i=1}^{1}\sum_{i=1}^{1} PH.\left(\overline{T}Cj + TCij.di.Yij + PH.HCj\right) + (PH.\ HCj) + (Dj.OCj) + HCj + \left(CKj + Bj2.\ Pk.\ HCj\right) + HCj.Z1 - a.LTj.Vj + \frac{\sum_{k}^{n} = 1\left[(Ckj+Bj).(Ckj+Bj)\right].p^k}{2\sqrt[2]{\frac{2Dj.OCj+HCj.B_j^2+\sum_{k=1}^{K}\left[Ckj+Bj).(Ckj+Bj).p^k\right]}{HCj}}} \tag{23}$$

$$C1 \le C2 \le \ldots \le Ck \le SS \tag{24}$$

$$Ckj \le HCj.Z<\text{ce:inf}>1\text{-a.}</\text{ce:inf}>\sqrt{LTj}.\sqrt{Vj}. \tag{25}$$

$$C_{2j\ge 0} \tag{26}$$

$$\sum_{j=1}^{J} Yij = 1 \tag{27}$$

$$\sum_{i=1}^{J} di.s.Yij \le capj.xj \tag{28}$$

$$\sum_{i=0}^{J} di.Yij = Dj \tag{29}$$

$$\sum_{i=0}^{J} vi.Yij = Vj \tag{30}$$

$$Xj \le 1 \tag{31}$$

$$Xj; Yij\varepsilon\{0,1\} \tag{32}$$

Eq. (24) ensures that the critical levels of determination are not greater than the level of inventory. Eq. (25) states that the K-grade customer has a higher critical level value than the rest, and the lower the customer grade (high priority customers), the lower the critical level

value. This means that each customer receives an order from only one warehouse. Eq. (26) $C2_j$: Critical level for customers with priority 2 in warehouse j. Eq. (28) indicates the capacity of the warehouse. Eqs. (29), (30) calculate the demand allocated to the warehouse and the variance of the demands. Eq. (31) also ensures that only one warehouse is established at each potential location.

3. Seeker evolutionary algorithm

Seeker Evolutionary Algorithm (SEA) is a new algorithm developed by Poursafary et al. (2014). Based on this method, search is regular and purposeful. In this algorithm, each answer is a member of the search group. At each stage, each solution searches in the part of the search space. After analyzing the information obtained from all search team members, the best parts of the search area are determined for the next step. The selected sections are divided into smaller sections in the next step, and the search team explore for them. The same process will continue until the stop condition is met. If the conditions are restarted, the algorithm returns to the first step, and in fact the search operation is performed again. These conditions can be the absence of expected improvement or reaching a certain number of repetitions. For example, repeat every 20 repetitions of the search operation or if the improvement was less than 0.0001. Of course, the best-case scenario for a resumption condition may be that there is no improvement in the number of iterations. The flowchart of the evolutionary search algorithm is shown in Fig. 2.

After the search operation, the spaces are ranked based on the results of the search team. In the first iteration, the right area had a better result than the left area. Among the parts, the top two parts are considered as selected parts. At this stage, because the total number of parts is equal to two, these two areas are selected as the selected area. In the second iteration, each of the selected areas is divided into two smaller parts. There are now 4 search areas. Therefore, 3 checkers are assigned to each section. The search operation is performed, and we rank the sections based on the results. We consider the top 2 areas as the preferred areas and continue this process until the stop condition is met (Cao et al., 2021). At any other iterations, if the condition is restarted, the algorithm will return to the first iteration. This structure creates a balanced process of convergence towards the optimal solution. It should be noted that in many algorithms, the speed of convergence of the solutions is very high, making it difficult for the algorithms to move towards the optimal solutions. However, it can be expected from SEA to converge to the optimal solution rapidly (Poursafary et al., 2014). Because it has two main and secondary loops.

- The main loop involves selecting search areas, dividing them, and assigning search team to each area.

Some of the parameters used in this algorithm are as follows.

nc_0:	Number of selected areas
na_0:	The number of smaller areas obtained from each selected area

These two parameters must be set for different problems. Usually, a value of 2 is good for these two parameters. By increasing the value of these two parameters, the probability of losing the area where there is an optimal value will be greatly reduced. The speed of convergence of the solutions will also decrease. In addition, the number of evaluations of the objective function will increase and, naturally, the solution time will also increase. Next, the following parameters are defined and used.

Υ: Minimum number of iterations to restart

β: Minimal improvement of objective function to restart

α: At least the size of an area to be divided into smaller areas nv: Number of variables to be shredded

These parameters are for restart conditions. For example, if $\Upsilon = 5$ and $\beta = 0.001$ are considered. In multiples of 5, if the algorithm performance rate is less than 0.001, the restart operation is performed. B can be considered to decrease after each restart. These speeds up the convergence of the algorithm towards the optimal point. For some problems, having a restart improves the algorithm's performance, and for some problems, its absence will result in a better result, for which a value of 8 times the maximum number of repetitions or more is set (Cao et al., 2021; Poursafary et al., 2014). If the chosen area was very small and the quantity is less α cannot divide the area into smaller parts. Because even in small sizes without division, the algorithm can achieve good results. This reduces the solution time and optimizes the structure of the algorithm. In a few problems, this parameter is used and in fact it takes on a value, and to solve most problems, the value of this parameter is zero. To divide the selected areas, the size of some variables must be divided. The criterion for selecting variables will be based on the size of the distance between their upper and lower limits, and its number is nv (Cao et al., 2021; Poursafary et al., 2014). The results obtained from various examples show that the most desirable value for the nv parameter is 1 or half or the total number of variables. In most optimization problems, despite a limit of 1, it is suitable for this parameter. The following parameters are defined and used for creating the stopping criteria.

$maxiter_0$:	Maximum number of iterations of the main loop
$maxiter_1$:	Maximum number of iterations of the sub-loop
$Best\ Seeker_0$:	The best solution found in the whole algorithm
$Best\ Seeker_1$:	The best solution found in the sub-loop of the algorithm

One of the stop conditions for the main and secondary loop of the algorithm is a certain number of iterations. In this algorithm, the maximum number of iterations of the whole algorithm is maxiter0 × maxiter1 × na0 × nc0.

In the SEA algorithm, the movement of search engines in their search area (sub-loop of the algorithm) is based on Eq. (33).

$$S^{new} = \left(w_1 * S^{old}\right) + (w_2 * Rand\ () * (Best\ Seeker)1 - S^{old}) \tag{33}$$

Each searcher moves to the best searcher of its search group at each stage, which also has a deviation in its path. Creating this deviation is obtained by multiplying a random number by the number of variables in the distance between the searcher and the best searcher. The process of this movement is shown in Fig. 3.

In order to find the movement for each seeker, w1 and w2 are defined as the weight of new seeker and best seeker respectively. If the values of w1 and w2 are considered equal to 1, the new position of our searcher will be between his old position and the best searcher with a deviation value. Search engine movements cause them to converge towards the best answer found. The results obtained from various examples show that the most optimal value for w1 is 0.5 w2 to 2. For some problems, there may be other suitable values that can be obtained by parameterizing these values. According to Eq. (33), moving the searcher causes the new status of the searcher in some variables to be outside the acceptable range for that variable. In this case, for those variables, the new situation is on the acceptable area boundary. If this happens to all variables, it means that the new browser status is completely out of the acceptable range. In this case, if the new condition is equal to the lower limit of the desired area (Varmin) of the specified area, Eq. (34) is used.

$$S^{new} = Var_{min} + Rand()(Best\ Solution - Var_{min} \tag{34}$$

And if the new condition is equal to the upper limit of the desired area (Varmax) of the specified area, Eq. (35) is used. And if the new condition is equal to the upper limit of the desired area (Varmax) of the specified area, Eq. (35) is used.

$$S^{new} = Var_{max} + Rand()(Var_{max} - Best\ Solution \tag{35}$$

The w1 and w2 values should be chosen so that the number of times the searchers are 310 completely outside the acceptable range is very small. The lower the value, the better the solutions. The maximum number of times this occurs should not be more than 25% of all cases (Cao et al., 2021; Poursafary et al., 2014). In the sub-loop of the sub-algorithm, we give searchers whose performance value is not appropriate a second chance to improve their performance. To do this, the

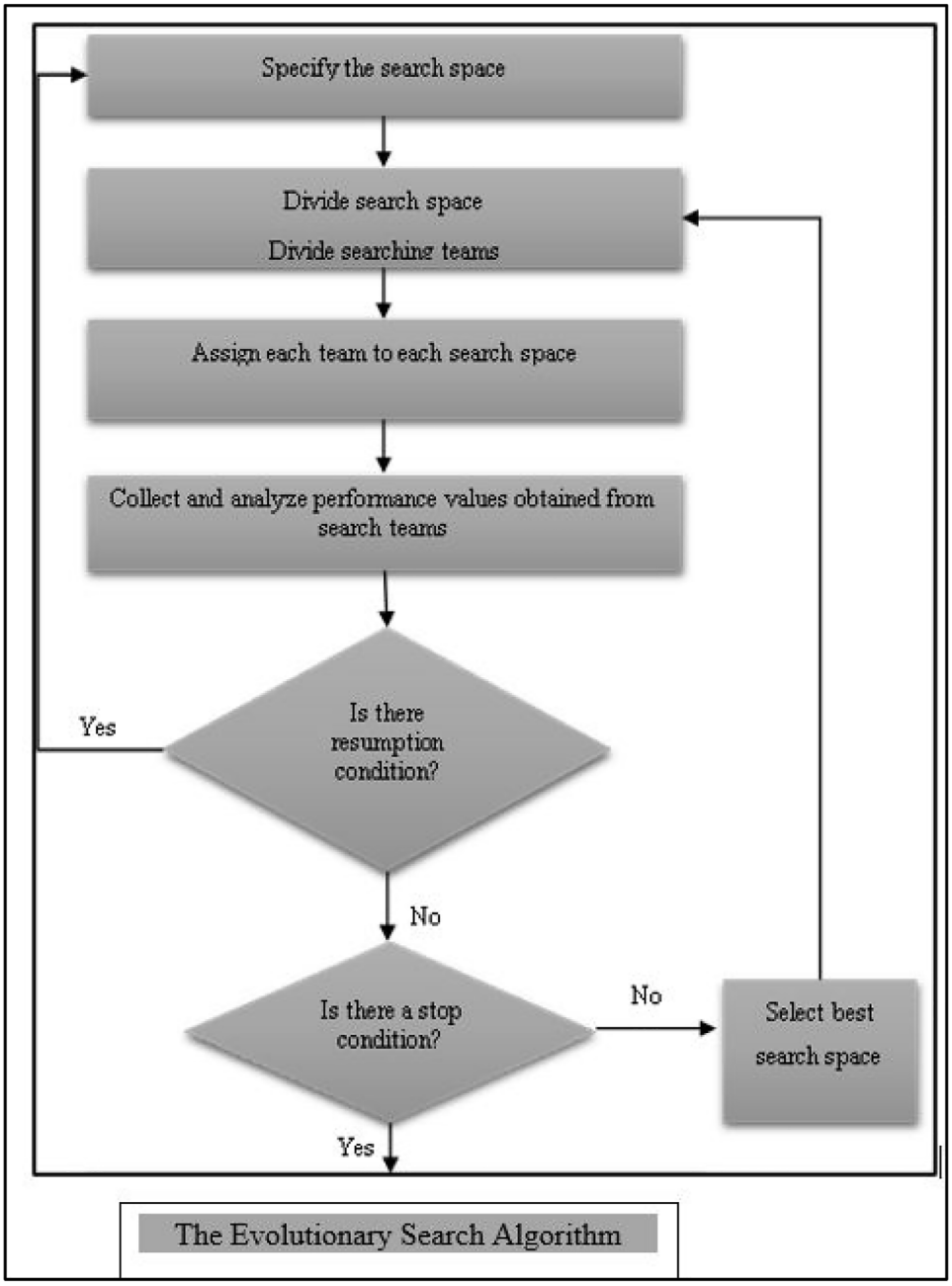

Fig. 2. The flowchart of SEA (Dai et al., 2006).

SEO performance values must first be normalized. This is done based on Eq. (36).

$$\left|fitness_i - Min\{fitness\}\right| \tag{36}$$

$$\sigma_i|Max\{fitness\} - Min\{fitness\}|$$

If the normalized value of a browser's performance (σ_i) is less than our random number between 0 and 1, that browser has a second chance to improve its performance by moving again according to Eq. (33). If the performance of that browser improves after the move, the status and performance of that browser will be updated (Cao et al., 2021; Poursafary et al., 2014). Of course, the maximum number of times this will happen should be a maximum of 25% of the number of searchers in each step. The pseudo-code of main loop of the proposed SEA is shown in Table 1 and the pseudo-code of SEA in each iteration is shown in Table 2.

Table 1
The pseudo-code of the main loop of the proposed SEA.

	Main loop
Step 1:	Specify the solution space as the selected area
Step 2:	Divide the selected area into *na0* sections
Step 3:	Assign search agents to areas and perform search operations (sub-loop)
Step 4:	Select *nc0* top area as selected areas
Step 5:	If the restart conditions are met, go to the first step
Step 6:	If the stop conditions are met, end the operation of the algorithm, otherwise return to the second step

4. Numerical results

In this section, the computational results obtained from the application of designed algorithms on different test problems are presented. Also, in order SEA algorithms, the results obtained using this algorithm

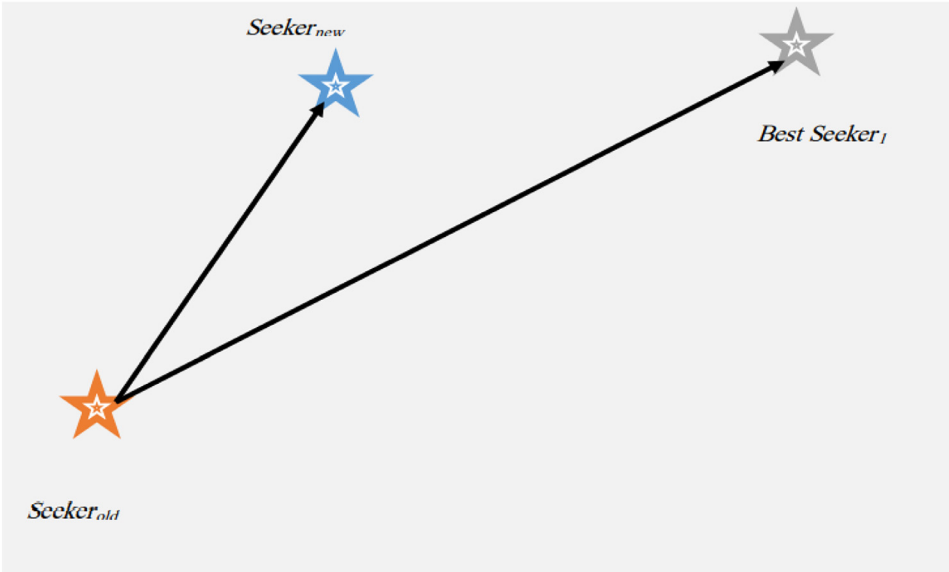

Fig. 3. Seeker move towards the best searcher (Poursafary et al., 2014).

Table 2
The pseudo-code of the sub-loop of the proposed SEA.

	Sub-loop
Step 1:	Generate random solution and calculate the suitability of each of them
Step 2:	If the solution fitness was better than Best fitness$_1$. Best solution$_1$ and Best fitness$_1$ are updated
Step 3:	Move each solution according to Eq. (35) and calculate their fitness
Step 4:	Repeat the second step
Step 5:	Normalize the fitness of the solution
Step 6:	If *rand* () $\leq \sigma_i$ Give every searcher a chance again
Step 7:	If the stop conditions were met, end the operation, otherwise return to the third step
Step 8:	If Best fitness$_1$ was better than Best fitness$_0$. Best fitness$_0$ and Best solution$_0$ are updated.

compared with genetic algorithm (GA) which is one of famous meta-heuristic algorithm and GAMS software. For this purpose, a certain number of problem examples are defined as follows: for each problem, the solution process is repeated 5 times and based on the results, the average of the objective function and time is presented.

4.1. Validation of proposed model and algorithm

In order to validate the proposed algorithms, 10 small test problems which are generated and shown in Table 3. These test problems are optimized using meta-heuristic algorithms and also compared with the solution obtained from GAMS software.

Table 4 shows the results obtained are optimizing different test problems. For accurate resolution, a time limit of 3600 s is considered for GAMS software.

Mean: According to the quality and shutter time of the solutions obtained by meta-heuristic methods as well as GAMS, the average values of the GA and SEA solution express how close they are to the global optimal solution obtained from GAMS.

The results presented in Table 4 show that according to the quality of the solutions and solving time obtained through meta-heuristic methods as well as GAMS, it is observed that the solution of GA and SEA is close to the global optimal solution obtained from GAMS with an approximate gap. It can be concluded that the gap was not significant compared to the time it took to solve the problem and that the solutions were relatively valid. Fig. 4 shows the comparison of metaheuristic methods based on their gap and Fig. 5 shows the comparison of solution times.

Table 3
The generated test problems in small and medial scale.

#problems	#customers	#warehouses	#category of customers	Number of inte-ger variables	Number of pos-itive variables
1	7	4	2	30	20
2	7	4	3	30	22
3	7	5	2	40	25
4	7	5	3	40	30
5	8	5	3	45	30
6	8	5	4	45	35
7	8	6	3	50	40
8	9	6	4	60	45
9	10	7	4	75	50
10	11	8	5	96	60

To optimally solve the problems, Baron solver is used in GAMS software. The genetic algorithm and SEA are also developed using the Matlab software. Using a computer with a 2.7 GHz processor and a 4 GB memory, the generated codes are executed, and the results are obtained.

The results presented in Fig. 5 show that the algorithms presented in all problem tests had an error of less than 10%. The average error of the genetic algorithm is 8.52% and the average error of the SEA is 7.54%, which indicates the superiority of the SEA algorithm. Fig. 6 presents a comparison of the solution time of different methods, the results indicate that GAMS software has a much higher solution time than meta-heuristic methods, and therefore the very high operating speed of the genetic algorithm and SEA is well visible.

4.2. Solving in large scale

To solve in real dimensions, 14 test problems were generated and each problem was solved 5 times by GA and SEA separately. The results of large-scale problem solving are shown in Table 3. A summary of the results is also provided in Figs. 6 and 7.

As shown in Fig. 6, the answer obtained from the SEA algorithm is better than the genetic algorithm. Due to the solved problems, SEA has worked more efficiently than the genetic algorithm in almost all cases. In terms of cost, except for issues 2 to 5, the SEA algorithm has reduced costs more than the genetic algorithm in other issues.

Also, according to Fig. 7, it is concluded that the SEA algorithm has spent more time than the genetic algorithm. Therefore, it can be concluded that in terms of solution time, it can be said that the SEA al-

Table 4
Comparison of the solution methods in small and median scale.

GAMS			GA			SEA		
	TotalCost	CPUtime	TotalCost	CPUtime	GAP(%)	TotalCost	CPUtime	GAP%
1	269.00	1.16	271.39	2.95	0.90%	269.15	3.75	0.10%
2	288.82	12.79	292.58	3.14	1.30%	314.86	4.10	1.80%
3	323.85	34.58	354.63	3.32	9.50%	340.34	4.32	1.10%
4	348.30	96.75	376.46	3.40	8.10%	340.47	5.04	1.00%
5	350.42	348.69	384.08	3.46	9.60%	389.46	5.53	9.50%
6	374.85	623.96	386.80	3.50	3.20%	439.04	6.16	3.70%
7	435.72	945.22	451.99	3.58	3.70%	506.67	6.96	0.90%
8	519.41	1276.68	520.01	3.83	0.10%	581.88	7.10	5.40%
9	523.32	2163.96	559.39	4.39	6.90%	695.31	8.41	7.10%
10	544.49	3600.00	587.05	5.02	7.80%	761.54	9.64	2.40%
Mean	397.82	910.38	418.44	3.66	8.52%	428.10	6.15	7.54%

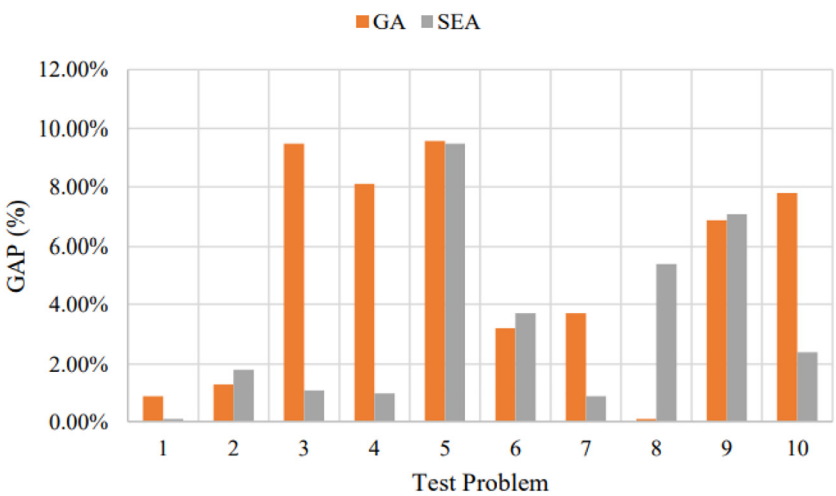

Fig. 4. Comparison of gap of meta-heuristic algorithms.

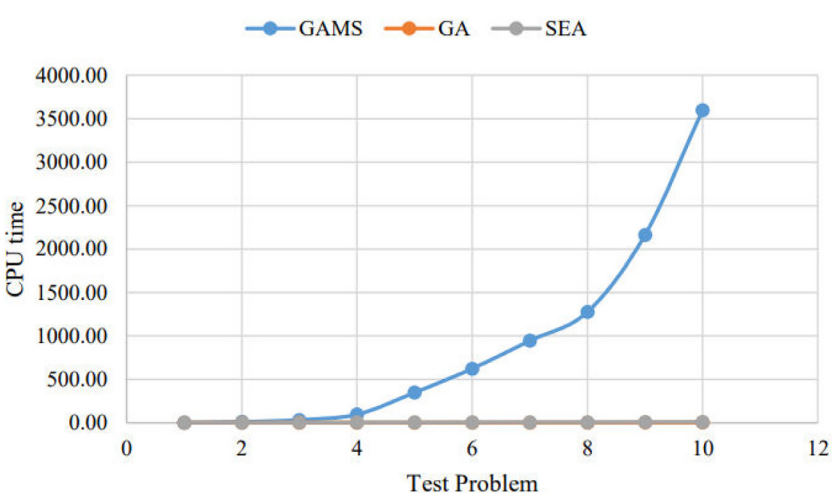

Fig. 5. Comparison of CPU time of different solution methods.

gorithm required more time than the genetic algorithm due to the existence of local search. But there is not much difference between the solving time of these two algorithms (Table 5).

4.3. Sensitivity analysis

In order to analyze the sensitivity of the proposed SEA algorithm to the problem parameters such as number of customers, potential warehouses, and number of customer groups, large-scale tests were performed on the test problem 1, in which the effect of increasing by constantly considering all model parameters One of the parameters is considered on the time of presenting the final answer to the problem.

As shown in Figs. 8 and 9, the increase in the number of potential warehouses in the problem has not affected the increase in total cost of supply chain. However, increasing the number of customers affect the CPU time significantly. Fig. 10 also shows that increasing the number of customer groups gradually increases the solution time.

Fig. 8 shows that increasing the number of customer categories increases the cost of the issue, which gradually reduces the cost slope. As can be seen in Figs. 9 and 10, the most influential and important factor in increasing the time to solve the number of customers. As seen from the chart above, with increasing number of customers, the solution time increases sharply. Also, according to Fig. 10, increasing the number of

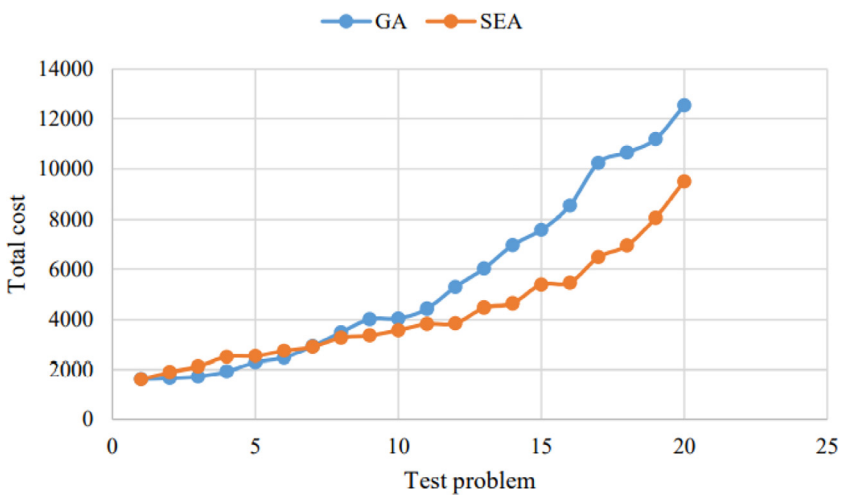

Fig. 6. Comparison of mean value of the objective function in the studied algorithms.

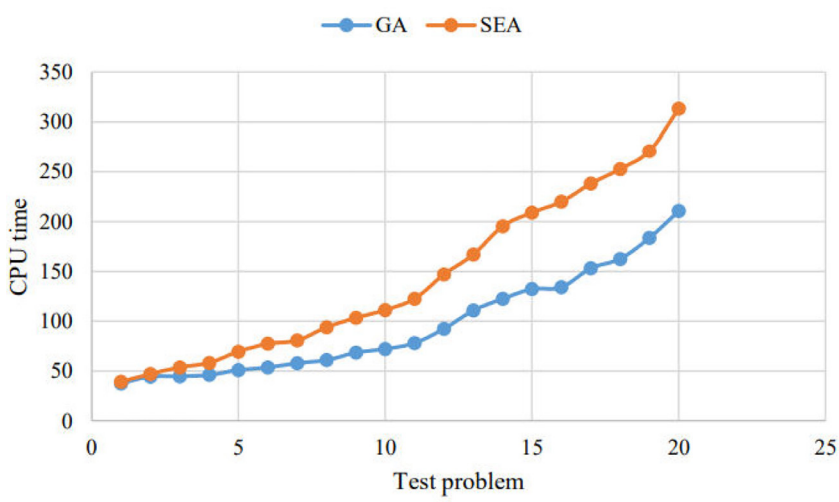

Fig. 7. Comparison of CPU time in the studied algorithms.

Table 5
Comparison of the solution methods in large scale.

Problem	GA				SEA			
	Total cost		CPU time		Total cost		CPU time	
	Mean	Max	Mean	Max	Mean	Max	Mean	Max
1	1621.21	1669.26	37.54	37.80	1610.82	1761.67	39.33	39.71
2	1654.791	1775.73	44.27	44.63	1884.85	2009.97	46.86	47.20
3	1723.066	1742.88	44.68	44.91	2124.76	2245.78	53.68	53.89
4	1918.238	1986.42	45.98	46.37	2517.13	2728.98	58.12	58.56
5	2286.338	2286.62	50.83	51.24	2549.86	2676.14	69.46	69.79
6	2481.074	2699.62	53.52	53.62	2753.50	2866.60	77.28	78.04
7	2944.308	3129.27	57.78	58.31	2920.81	3178.88	80.58	81.34
8	3501.351	3521.32	60.88	60.91	3283.92	3302.31	93.84	94.53
9	4022.708	4301.95	68.34	68.69	3372.71	3389.07	103.28	104.26
10	4049.13	4186.14	72.11	72.77	3583.41	3625.30	111.23	111.87
11	4425.923	4782.01	77.97	78.00	3832.10	4182.59	122.46	123.64
12	5289.203	5360.61	92.31	92.44	3854.94	4024.84	146.91	147.94
13	6025.12	6392.36	110.66	111.72	4473.59	4724.32	166.83	167.97
14	6955.573	6959.52	122.46	122.97	4642.63	4745.30	195.31	197.11

customers increases the cost of the issue and plays an important role in it.

5. Discussion

Optimal location of the warehouse ensures success of supply chains in terms of low cost and more profit. It becomes inevitable for the firm to lay stress on deciding the perfect location among various location alternatives present for setting up a warehouse. Thus the amount of investment done in setting up and developing a warehouse should always follow a close analysis directed towards finding out the most optimal locations amongst the various location alternatives present (Chorafas, 1974). In our research, we focused on the difference in the importance of customers to the central company. A study by Kar & Pani (2014) aimed to identify critical supplier selection criteria that are important across industries and purchasing contexts, and then estimate

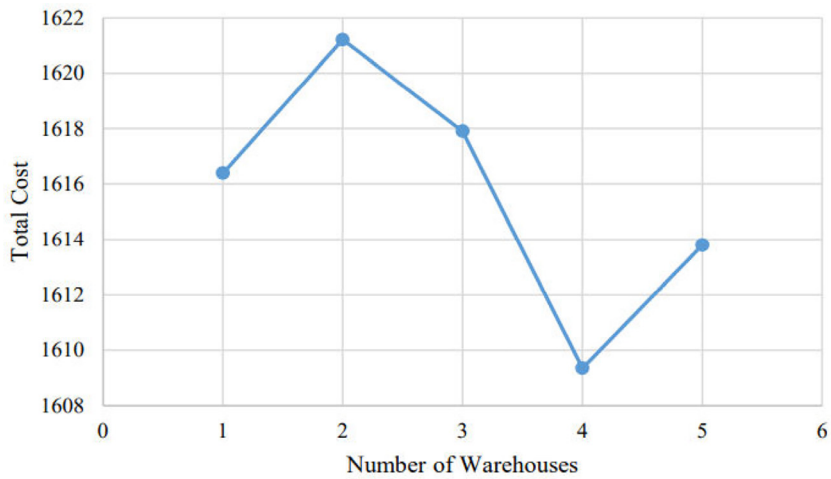

Fig. 8. The role of potential warehouse locations on the total cost of supply chain.

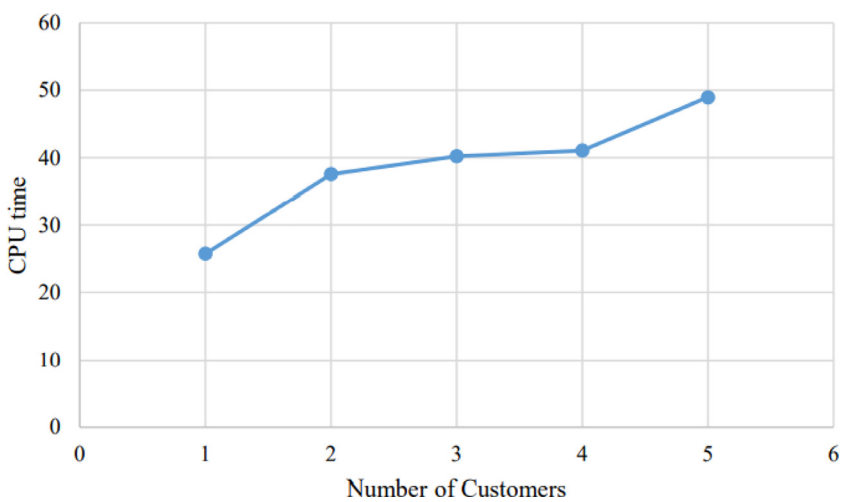

Fig. 9. The role number of customers on the SEA CPU time.

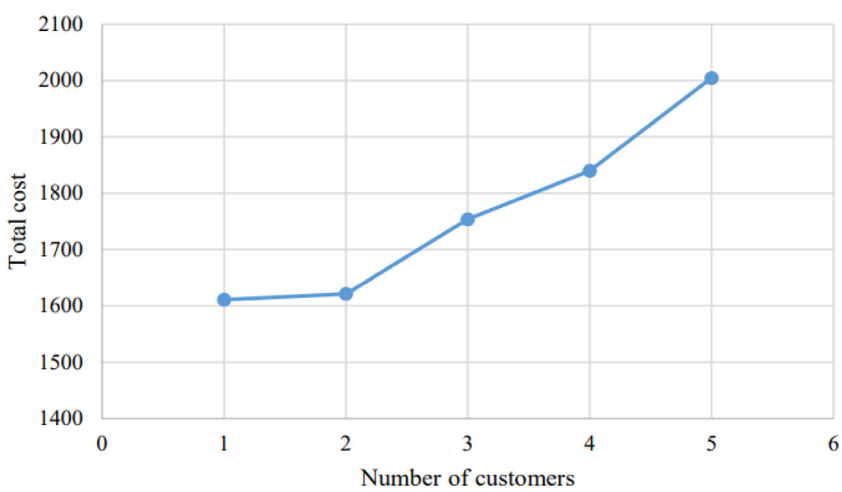

Fig. 10. Cost changes with increasing number of customer categories.

the importance of these criteria for procurement practitioners. Taking these differences into account, we have introduced allocation and inventory control and marketing policies. Compared to the several research items done in the field of inventory optimization, few ones have considered inventory with different customer classes (Melo et al., 2009). Deepu & Ravi (2021) research has introduced a new model for choosing the best IOIS alternative, taking into account the content, scope, and critical decision-making factors that affect supply chain integration. In this context, we aimed to provide an integrated model for location and inventory planning in the supply chain in our article in order to fill the gap in the literature. Wang et al. (2013) recognized two classes of customers as high priority L-class customers and low priority B-class customers due to lack of orders, delays and also loss of sales. Previous research on supply chain network design and customer classification has been reviewed and described in our study. Kaplan (1969) evaluated the inventory system with two classes of customers and a two-critical system with critical and

noncrit critical levels. Abdolazimi et al. (2021) examined the determination of the optimal inventory number of each product group in a supply chain and used ABC analysis in the study. In our article, we designed a complex and nonlinear integer programming model to solve hypotheticals using two genetic and SEA algorithms, caring that they exist real and actually exist and manage the problem area. Kar (2014) proposes an approach for group decision support for the supplier selection problem by integrating fuzzy Analytic Hierarchy Process (AHP) for group decision making and fuzzy goal programming for discriminant analysis. Optimizing the model we present in this article is not easily possible with precision methods due to its very high complexity, and the best approach is to use innovative approximate methods. Further research has concluded that the use of innovative methods such as SEA algorithms will not be very useful, since the objective function and some limitations of the proposed model are not linear. Finally, it was concluded that metaheuristic methods such as genetic algorithms and SEA algorithms would be the best choice to solve the model. Yoo & Cheong (2021) emphasized the application of the Seeker Evolutionary Algorithm as a new and efficient meta-intuitive algorithm in the field of supply chain network design in the study, which examined the stability of the closed loop supply chain based on inventory behavior. In our study, we found that solving time using Meta intuitive algorithms ranges from 3 to 316 s and is a very good time compared to GAMS software. Kaplan (1969), Zhou & Zhao (2010), Abdolazimi et al. (2021), Yoo & cheong (2021) is not offered directly to the customer and is shipped at customer's request after being stored in warehouses. In our article, we have proven that meta-intuitive algorithms offer good and valid solutions in a reasonable time by comparing solutions obtained from small-scale problem solving with precise solution methods. In addition, it has been found that increasing the solution time by drawing the solution time in order to increase the dimensions of the problem is polynomial order, that is, it means the high efficiency of the algorithms used. Finally, an analysis of the resolution time and cost sensitivity of the supply chain was conducted, indicating that increasing the number of customers and the customer category would have the greatest impact on the solution time and total costs of the supply chain.

6. Conclusions and suggestions

Allocation and inventory control and marketing policies were presented taking into account the difference in the importance of customers for the central company. Accordingly, this paper presents an integrated model for location and inventory planning in the supply chain. The problem was defined and explained by reviewing previous research in supply chain network design and customer classification. It was also tried to consider the assumptions as real and as they exist in reality and govern the problem space, and finally a complex and nonlinear integer programming model was designed to solve it using two genetic and SEA algorithms. Optimizing the model presented in this paper is not easily possible through precise methods due to its very high complexity, and the best approach is to use innovative approximate methods. Further investigation concluded that the use of innovative methods such as SEA algorithms would not be very useful because the objective function and some of the constraints of the proposed model are nonlinear. Finally, it was concluded that metaheuristic methods such as genetic algorithms and SEA algorithms will be the best choice to solve the model. Solving time using meta-heuristic algorithms varies from 3 to 316 s, which is a very good time compared to the GAMS software. Moreover, by comparing the solutions obtained from small-scale problem solving and comparing with exact solution methods, it was proved that metaheuristic algorithms provide good and valid solutions in a reasonable time. Also, by plotting the solution time in terms of increasing the dimensions of the problem, it was found that increasing the solution time is a polynomial order, which means the high efficiency of the algorithms used. Finally, the analysis of the sensitivity of the solution time and cost of the supply chain was performed, which showed that increasing the number of customers and the category of customers will have the greatest impact on the solution time and total costs of supply chain. With the model developed by the enterprise, supply chain weakness is eliminated, but also eliminates human investment in the supply chain, which is a weakness. As a result of all this, it is possible for the business to have a growth opportunity because the competitive structure increases. In terms of eliminating these risks by separating the risks, it is important for enterprises to have time to observe the sector, not to cover their own weaknesses. Weaknesses in the supply chain are eliminated with the developed model. As a result, businesses are likely to stand out by differentiation. The program, which was developed today because there are businesses that still have a problem on the Internet, whose whole system collapses and does not care about costs, has the ability to offer a broader perspective to these businesses. In future research, general metaheuristic algorithms or combinatorial metaheuristic algorithms can be expanded.

Author contributions

Each of the authors were responsible for their sections as follows.

Conceived and designed the study ESD; methodology; software, ESD; EA, validation, MBÖ; EA formal analysis, ESD performed the research; investigation, ESD, EA, MBÖ resources, ESD, EA, MBÖ data curation, ESD; writingoriginal draft preparation, ESD, EA, MBÖ writingreview and editing, ESD, EA, MBÖ supervision, ESD, EA analyzed the data and MBÖ contributed to editorial input.

All authors have read and agreed to the published version of the manuscript." All authors have read and agreed to the published version of the manuscript."

Declaration of Competing Interest

The authors declare that there is no conflict of interests regarding the publication of this paper.

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